

Trust-aware resilience for connected vehicle platoons

From attack-aware state estimation
to an onboard edge module

Project overview for potential users and
decision-makers

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Connectivity creates physical risk

CONTEXT

Cooperative driving depends on data the vehicle cannot fully trust.

What connectivity enables

- Shared position, speed, heading and acceleration
- Cooperative cruise control and coordinated platoons
- Smoother traffic flow with fewer onboard measurements

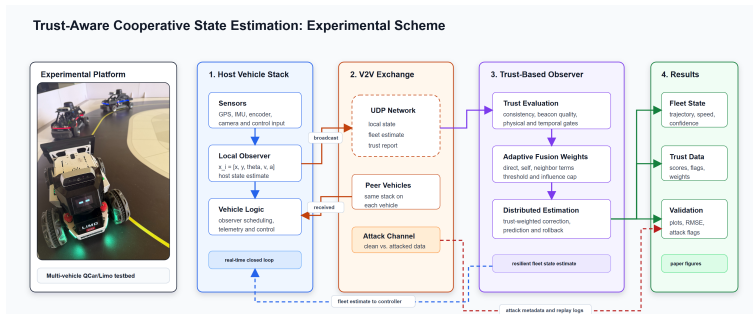
What an attack can change

- False, replayed, delayed or missing V2V packets
- Corrupted estimates that propagate across the fleet
- Unsafe control decisions based on plausible-looking data

Preserve cooperative value; degrade safely when V2V evidence is unreliable.

A trust layer protects cooperation

PROJECT CONCEPT



The trust layer screens, weighs and corrects data before the existing controller.

The vehicle ECU retains command authority

ELECTRONIC BOARD INTEGRATION

Inputs

- IMU and local kinematics
- GNSS position and 1PPS timing
- Vehicle-bus data and V2V packets: Wi-Fi now, C-V2X later



Edge functions

- Time and packet-validity gates
- State estimation, trust and source weighting
- Rollback, re-anchoring and audit logs



Outputs

- Trusted state estimates
- Attack, trust, availability and health flags
- Safe-mode request via CAN/CAN-FD gateway

Planned path: trust flag → gateway → ACC/CACC ECU → safe-mode request. The ECU retains acceleration and braking authority; deployment begins monitor-only.

Intervention is graduated, not binary

RESPONSE WHEN AN ATTACK IS DETECTED



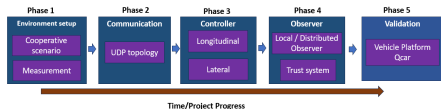
Two complementary effects: rejection stops future influence; rollback and anchoring revise recent contamination.

Current scope: the prototype does not claim direct actuator override; the vehicle controller retains command authority.

Two tracks validate software and hardware

WORK PERFORMED: TWO VALIDATION TRACKS

Software / algorithm track

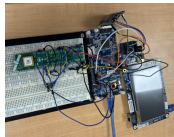


- Five-vehicle numerical attack campaign
- Q Labs plus QCar/LIMO packet exchange and closed-loop tests

Measured evidence: 100% detection in the tested campaign; 28–88 ms mean threshold-crossing delay.

One architecture, three roles: suppress unreliable inputs, repair delayed contamination and inform safe control degradation.

Hardware / embedded track



- Four-layer PCB with MCU/RTOS architecture
- IMU, GNSS timing, Wi-Fi and logging interfaces

Bring-up evidence: stable 100 Hz RTOS cycle, UART, GNSS acquisition and functional Wi-Fi 6 tests.

Trust suppresses unreliable sources

CORE METHOD - TRUST-AWARE DISTRIBUTED ESTIMATION

Reliability becomes an online fusion weight instead of a fixed assumption.

100%

Detection across all tested attacks and channel configurations

80.64–98.44%

Attacked-source zero rate during the attack window

28-88 ms

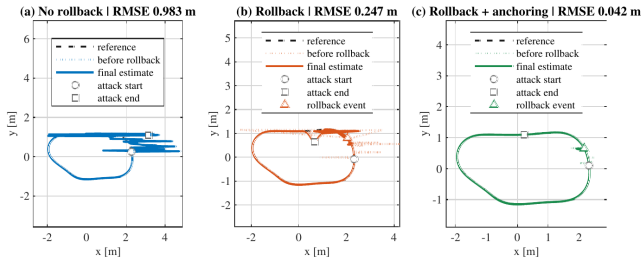
Mean threshold-crossing delay after the 10 s attack onset

Corrupted channel	Mean combined RMSE	Trust drop	Source zero rate
Local	0.643	0.319	80.64%
Global	0.393	0.460	92.90%
Local + global	0.567	0.544	98.44%

Rollback removes stored contamination

RECOVERY MECHANISM - DELAYED-DECISION ROLLBACK

XY trajectory comparison for target V0



No rollback
0.983 m

Rollback
0.247 m

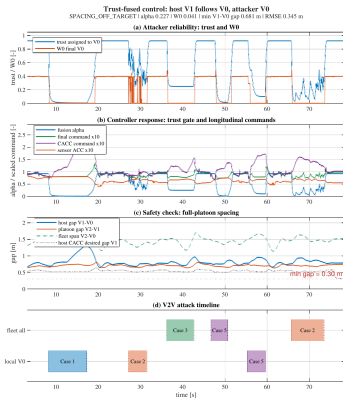
Rollback plus accepted
relative-pose anchoring
0.042 m

Local-channel Case 2 RMSE.

Correction reduces error 54.4%;
anchoring reduces it 75.2%.

Trust enables safe control degradation

CONTROL APPLICATION - TRUST-GATED ACC/CACC



< 0.05

Mean attacked-source observer weight

0.681 m

V1 minimum path-projected gap

0.637 m

V2 minimum path-projected gap

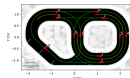
Both gaps remain above the study's 0.30 m reporting margin. The result is a feasibility check, not a formal full-footprint collision guarantee.

Tests reproduce the full operating loop

QUALITATIVE RESULTS



Physical QCar/LIMO scale-model track



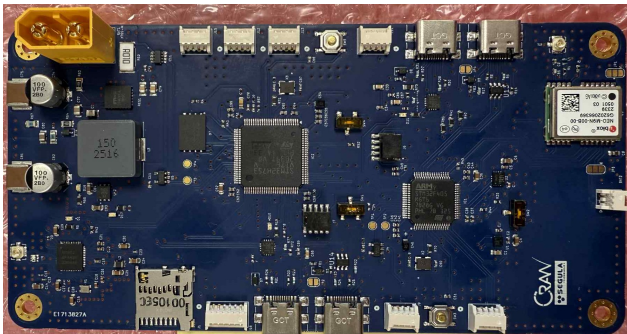
QLabs trajectory and waypoint scenario

One operating loop across both environments

Same state, communication and control concepts; real packet timing; repeatable attacks and logged recovery.

The deployment platform is tangible

MANUFACTURED ELECTRONIC BOARD



- Four-layer PCB with EMC-oriented grounding and power routing
- MCU/RTOS architecture targeting approximately 100 Hz estimation
- IMU, GNSS timing, Wi-Fi, storage and development interfaces

Bring-up evidence: stable RTOS operation at 100 Hz, UART, GNSS acquisition and functional Wi-Fi 6 tests.

Research novelty is not product readiness

STATE OF THE ART, COMMUNICATION AND INDUSTRIAL STANDARDS

The research contribution is functional integration; product readiness depends separately on interfaces, cybersecurity, safety and qualification.

Research position

- Trust-adaptive V2V fusion replaces a fixed-reliability assumption
- Suppression plus rollback/anchoring address future and stored contamination
- Nonlinear 2D platoon model with local and global channel attacks
- Trusted estimates are exposed to a control-availability decision

Industrial position

- V2X messages and CAN/CAN-FD provide interfaces; trust adds runtime data reliability
- ISO/SAE 21434, UNECE R155 and ISO 26262 processes still apply
- Secure boot, signed updates, key protection and protected logs remain required
- The research board still needs automotive EMC, environmental and vehicle qualification

Position: broader functional scope and modularity—not universal superiority; research-validated and hardware-demonstrated, but not automotive-qualified.

SEGULA can reuse this research

ECONOMIC PATH: REUSE RESEARCH ASSETS ACROSS CUSTOMER PROGRAMS

01

Reusable assets

Trust and rollback software, an embedded reference platform, repeatable attacks, digital-twin scenarios and synchronized logs.

02

Client work packages

V2X-resilience studies, HW/SW prototype integration, cyber-robustness campaigns and safety, EMC or vehicle-validation support.

03

Value to test

Less repeated setup, a differentiated OEM/Tier-1 offer and follow-on engineering from concept through validation and industrialization.

Commercial gate: confirm customer demand, integration effort, reuse rate and engineering time saved before making any ROI claim.

A shadow-mode pilot is the next gate

RECOMMENDED CONTINUATION

Three coordinated workstreams define the industrialization path.

Hardware

- Production board revision
- Automotive components, power and thermal design
- CAN/CAN-FD plus production V2X
- EMC and environmental qualification

Software

- Improve trust, rollback and anchoring
- Evaluate data-driven / AI augmentation
- Cover false positives, delay, dropout and stealth attacks
- Optimize execution time and memory

Safety and vehicle use

- Select a valuable cooperative-driving use case
- Integrate in shadow mode on the vehicle
- Validate detection, timing and fallback behavior
- Build ISO/SAE 21434 and ISO 26262 evidence

Decision gate: proceed with a monitor-only pilot when a vehicle program provides bus access, a measurable use case and owners for all three workstreams.